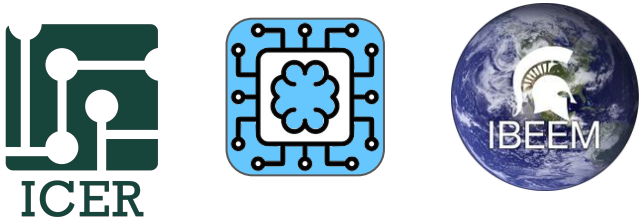
GenAI on the HPCC



Julian Venegas
Institute for Cyber-Enabled Research

Kelly Kapsar
Institute for Biodiversity, Ecology, Evolution, & Macrosystems

Introductions
&
Overview

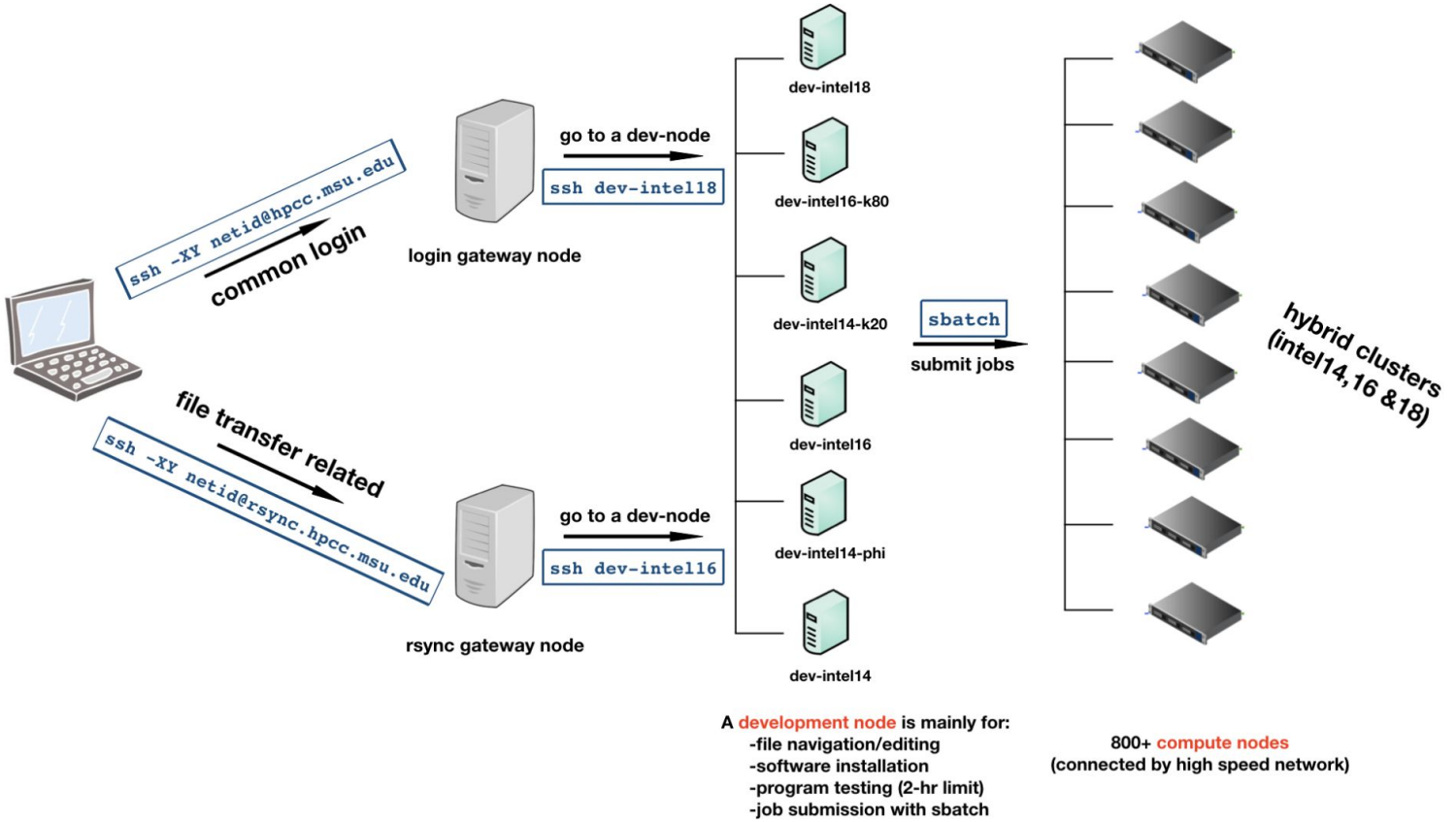
Institute for Cyber-Enabled Research

ICER provides:

- Free access to **high performance computing** platforms
- **User support**, and **consultation**
- Specialist staff experienced in scientific computing, machine learning / AI, across various domains



HPCC Layout



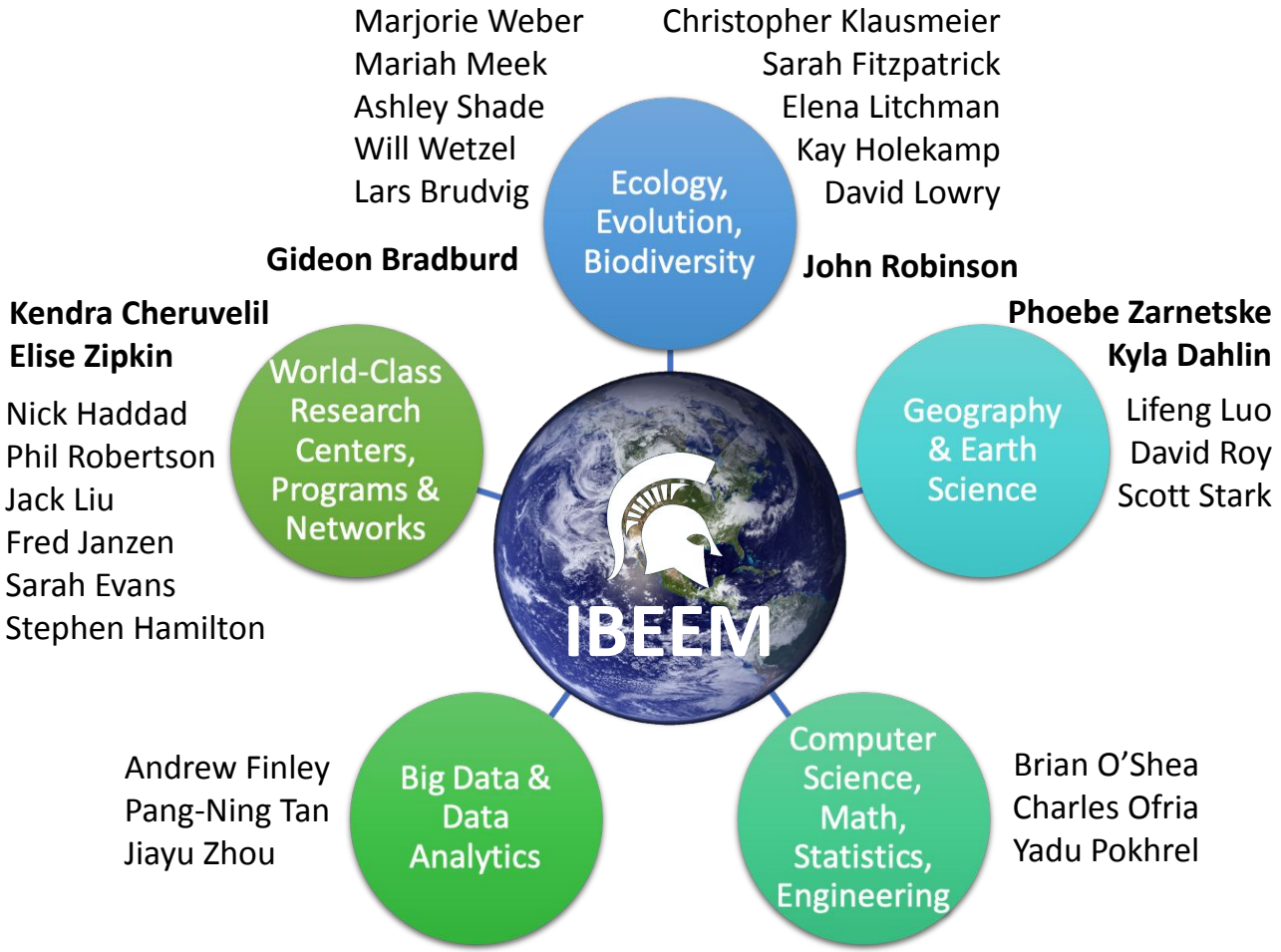
[https://docs.icer.msu.edu/HPC_s_entire_layout_at_ICER/]

Institute for Biodiversity, Ecology, Evolution, & Macrosystems

IBEEM Team

IBEEM increases:

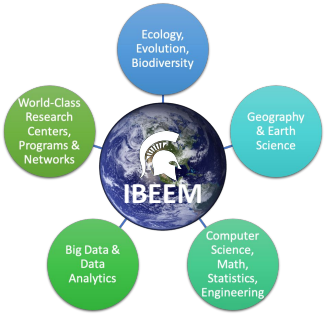
- *synergy*
- *capacity*
- *opportunities*
- *stature*



Institute for Biodiversity, Ecology, Evolution, & Macrosystems

Central Question:

What makes the biodiversity of a species, population, community, or ecosystem resilient to global changes?



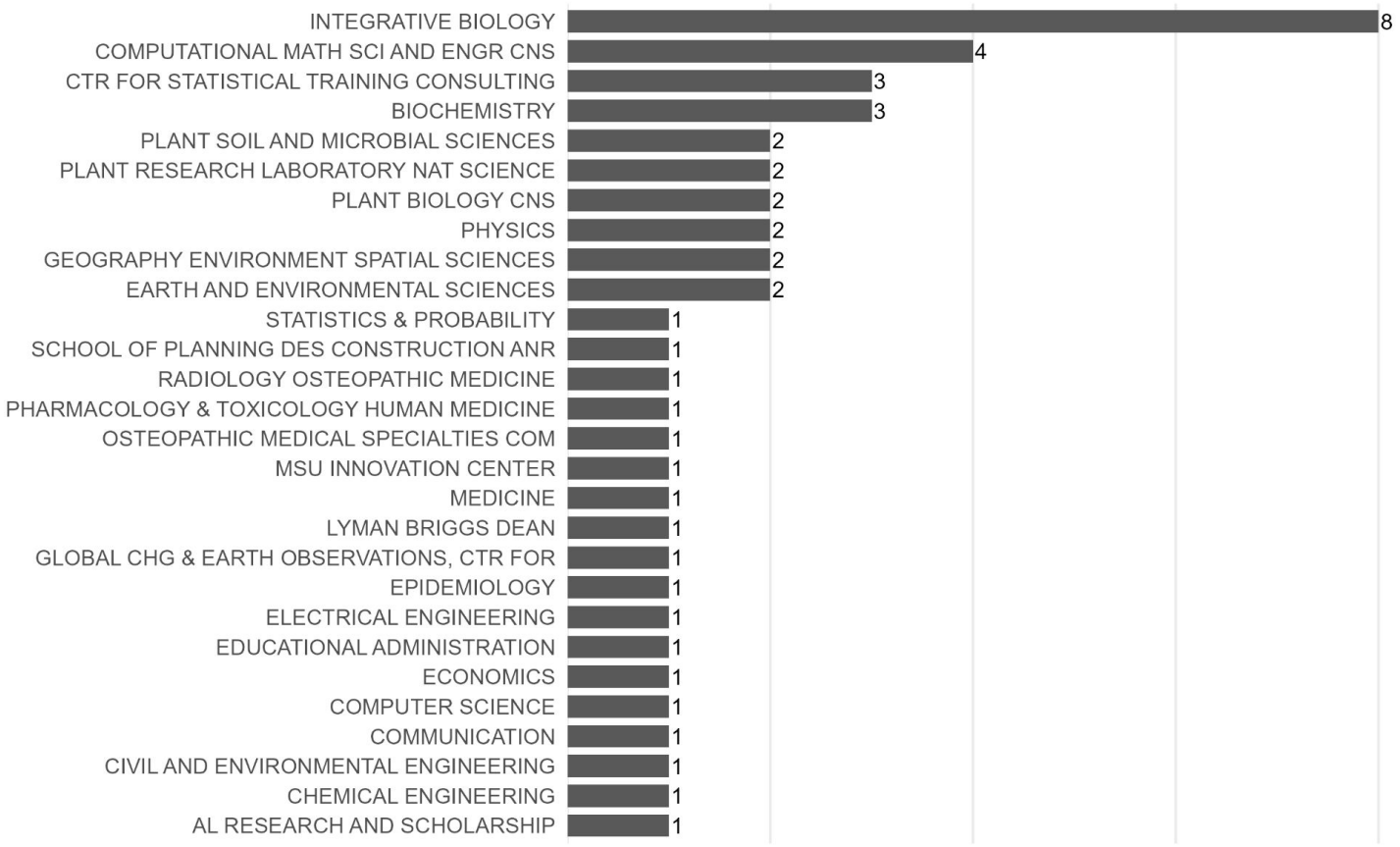
3 IBEEM Themes:

- Advancing biodiversity science with macrosystems
- From genes to ecosystems: Resilient biodiversity in the face of global change
- Harnessing the data revolution: forecasts for biodiversity responses to climate and land use changes

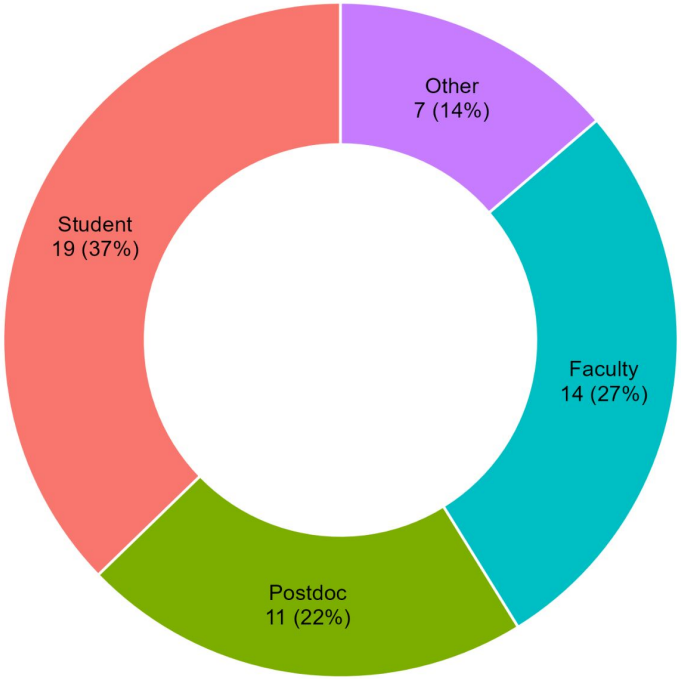




Attendee Breakdown



Attendee Breakdown



Outline

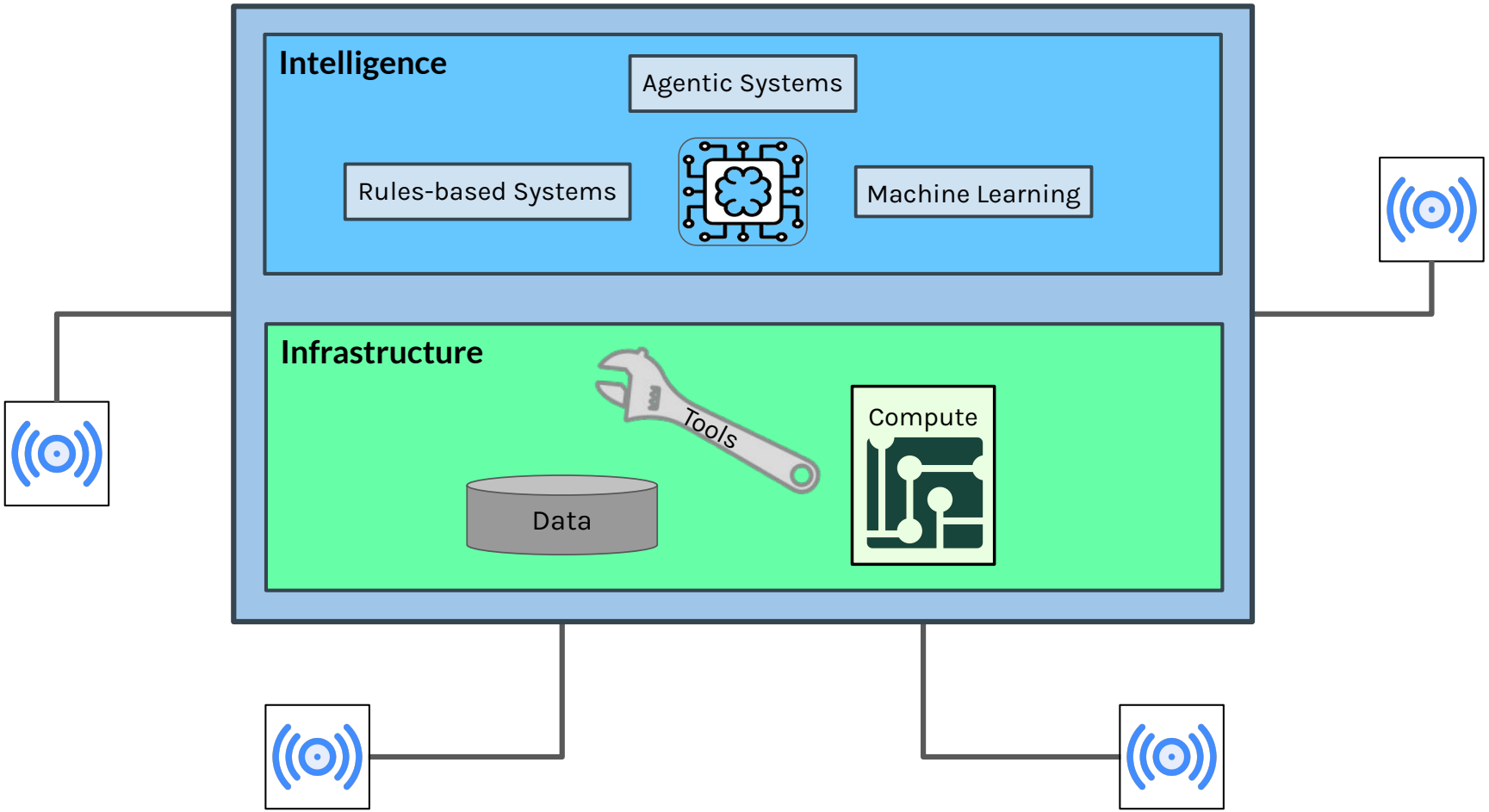
- AI fundamentals
  - Intro
  - Feature engineering vs. feature extraction
  - Model learning types
  - Fine-tuning models
  - Retrieval augmented generation (RAG) models

Break

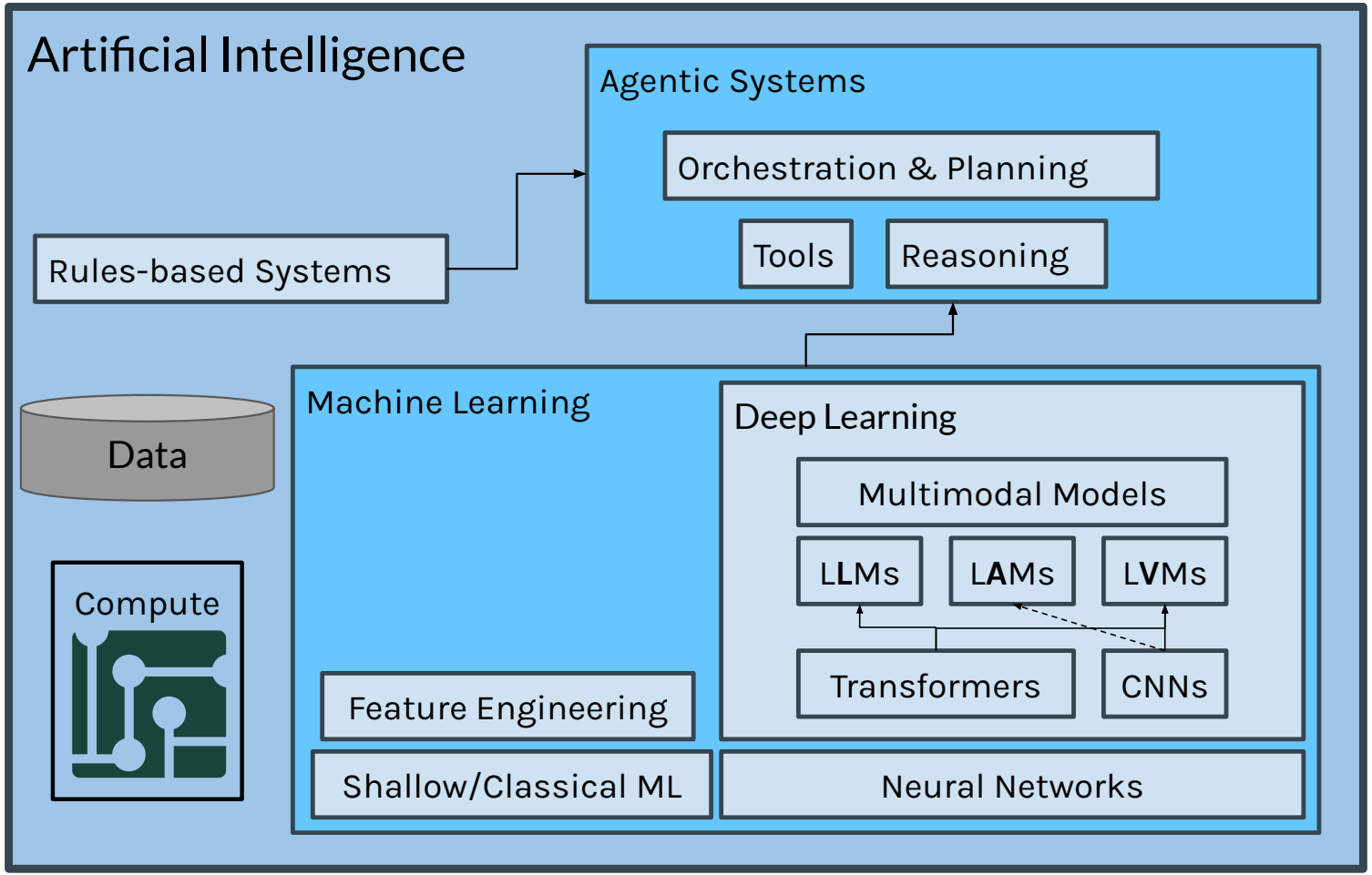
- MSU AI Policy/Guidelines
- AI tools on the HPCC
  - HPCC OnDemand: OpenWebUI and LM Studio
  - Ollama
- Use cases: chatbots (**highlight unstructured output**)

AI Fundamentals

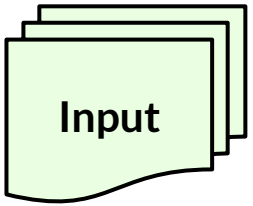
AI/ML Fundamentals



AI/ML Fundamentals



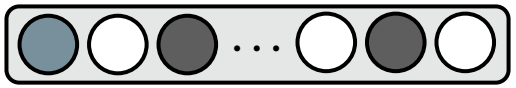
AI/ML Fundamentals



Numeric data
Words
Images
Audio

Satellite image
MRI
Text prompt

Features/Descriptors



A measurable variable (manually calculated or computer-generated)

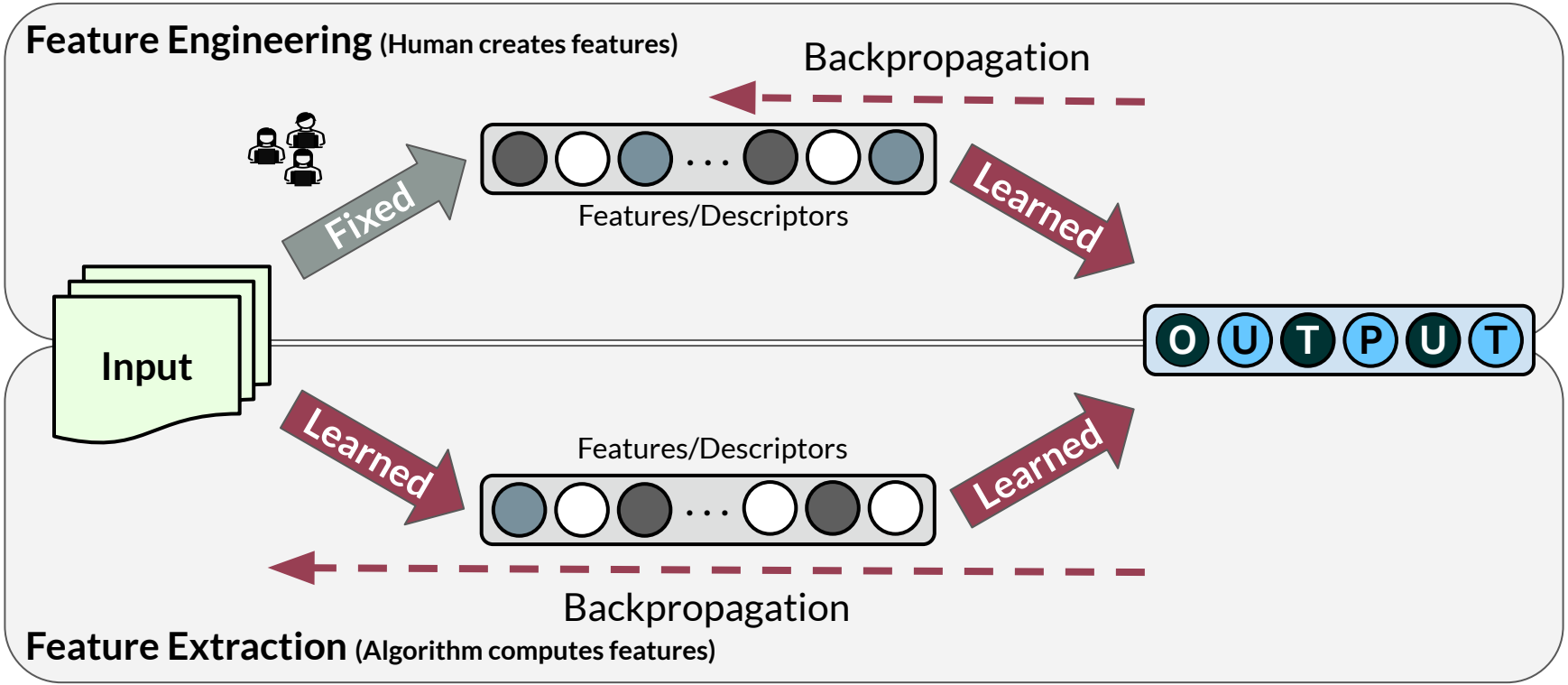
Vegetation greenness calculation
Edge detection
Embeddings



Can take a variety of forms (e.g., labels, clusters, annotations)

Landcover type
Tumor ID
Text response

Feature Engineering vs Feature Extraction



2-Layer Perceptron

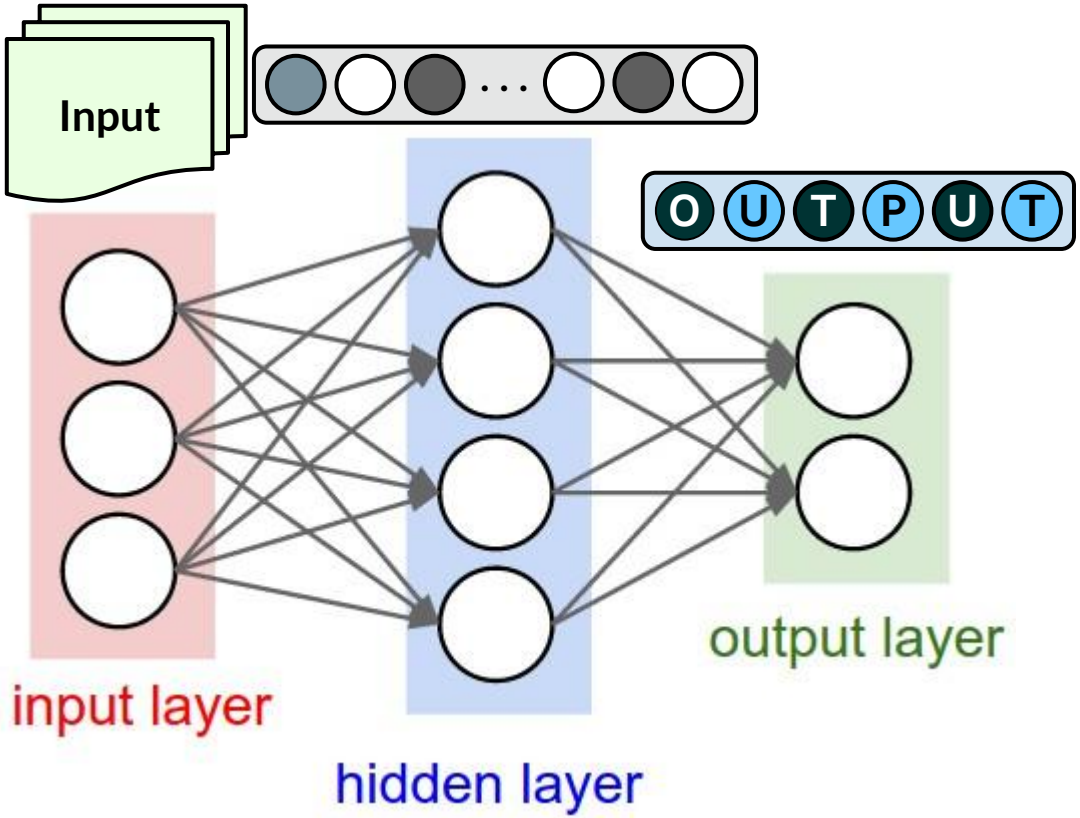
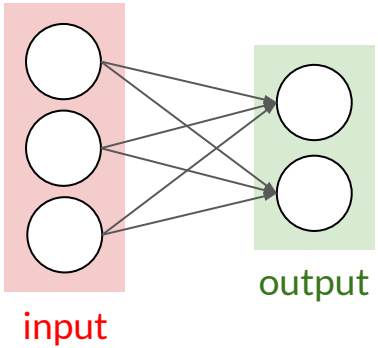
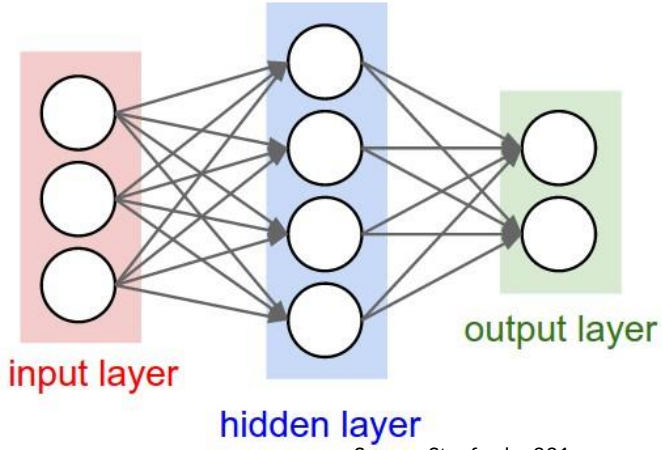


Figure from Stanford cs231n course <https://cs231n.github.io/>

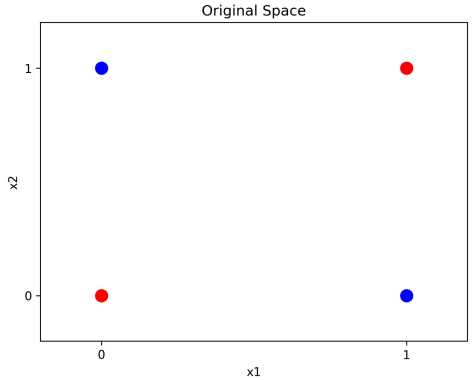
2-Layer Perceptron



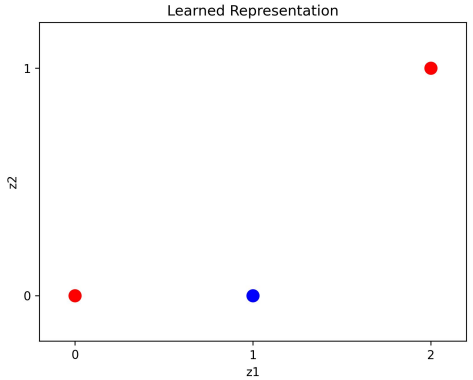


Source: Stanford cs231n course <https://cs231n.github.io/>

single-layer perceptron*



2-layer perceptron



XOR example: Goodfellow et al., Deep Learning (2016), Fig. 6.1

Supervised vs Unsupervised Learning

Supervised

- Model learns from labelled data
- Regression & Classification tasks
- Learn feature space to best predict labels
- Ex outputs:
  - Presence/absence of animals in a camera trap photo
- Example methods:
  - Regression based methods: linear regression, decision trees, gradient boosting, regression w/ NNs
  - Support-vector machines, Logistic regression, k-Nearest Neighbor, NNs

Unsupervised

- Model identifies structures/patterns from data without labels
- Clustering, Dimension Reduction, Generative Models
- Learn feature space to capture underlying structure and/or distributions
- Ex outputs:
  - Clusters of similar pixels in a satellite image or MRI scan
  - Most common topics in a newspaper archive
- Example methods:
  - Principal Components Analysis
  - K-means clustering

Supervised vs Unsupervised Learning









Self-Supervised Learning



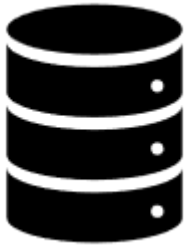
- Supervised style training w/ pseudo labels
- Pretext tasks (next token pred, predict missing image patches)
- Representation learning for downstream tasks.

Structured vs Unstructured Tasks/Output



Structured

- Pre-defined, fixed formatting requirements
- Useful for when output of an AI/ML model is one part of a larger pipeline/workflow
- E.g, entity-relationship extraction, tables (CSV), JSON, categories, labels



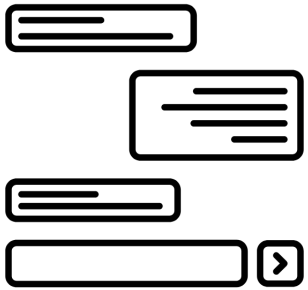
S1	S2	Eff1	Eff2	Type	Conf	...
e	f	-1	1	Pred.	0.9	...
g	h	1	-1	Par.	0.85	
g	k	-1	-1	Comp.	0.5	...

Unstructured

- No fixed format
- Easily human-readable
- Ex: freeform text, Q&A/chat, summaries



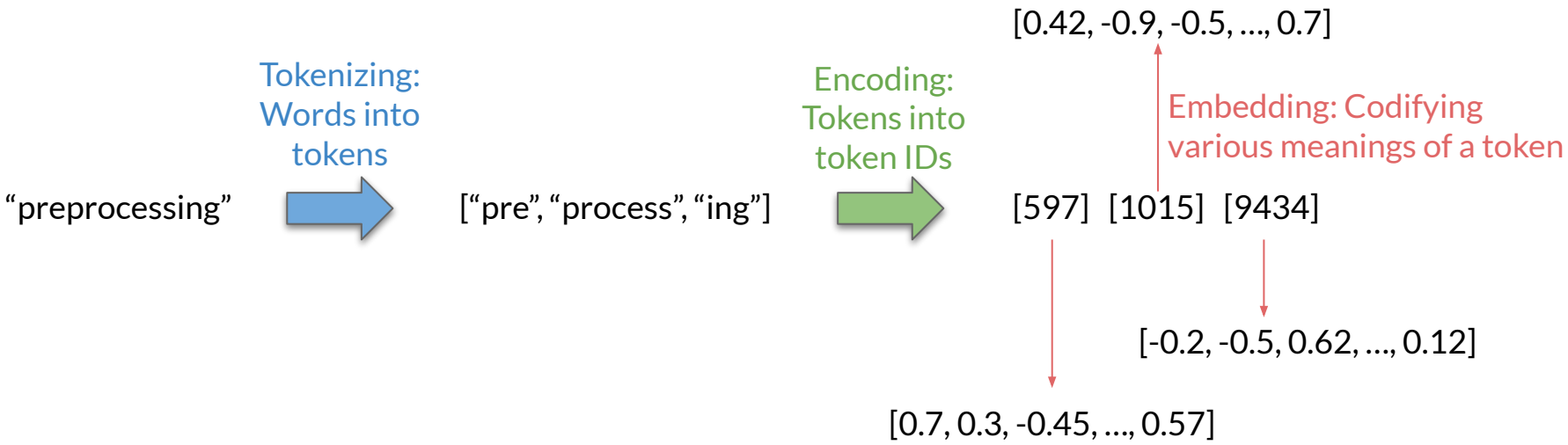
Created by Agan24
from Noun Project



Created by Gonza Monta
from Noun Project

Tokenization, Encoding, and Embedding

- The process of breaking down text into the smallest useful units and assigning values to those units.
- E.g., Word-level (intuitive), Byte-pair encoding



Reference:
https://medium.com/@mohitgupta.mg878/from-raw-text-to-embeddings-the-complete-llm-input-pipeline-2706cc5a2e37

Next Token Prediction

- Autoregressive
- Unidirectional
- Probabilities influenced in part by temperature setting
- Decoder-only, generative models
- Form of self-supervised learning

Enter text:

One, two, three,



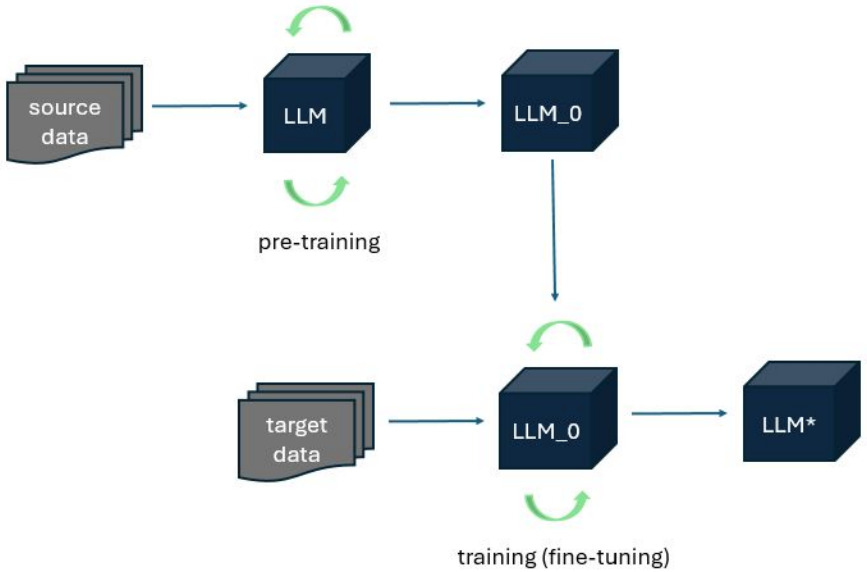
3198 11 734 11 1115 11

Prediction

#	probs	next token ID	predicted next token
0	46.44%	1440	four
1	7.48%	290	and
2	7.31%	1936	five
3	2.66%	393	or
4	2.54%	2237	six
5	2.09%	1115	three
6	1.86%	3863	maybe
7	1.62%	345	you
8	1.23%	257	a
9	0.92%	530	one

Image & example from < https://huggingface.co/blog/alonsosilva/nexttokenprediction >

AI/ML Fundamentals: Fine-tuning



Foundational Model Pipeline

0. Pre-training



- Input: Massive corpus
- Process: Often self-supervised (e.g., next token prediction)
- Output: Base model

1. Supervised fine-tuning

- Process:
  - Smaller, labeled data
  - Targeted behaviour, tasks/domains
  - Align w/ human judgement
- Output: supervised fine-tuning (SFT) model

2. Reinforcement learning from human feedback

- Align w/ human judgement
- Reward for human preference

3. Human guardrailing

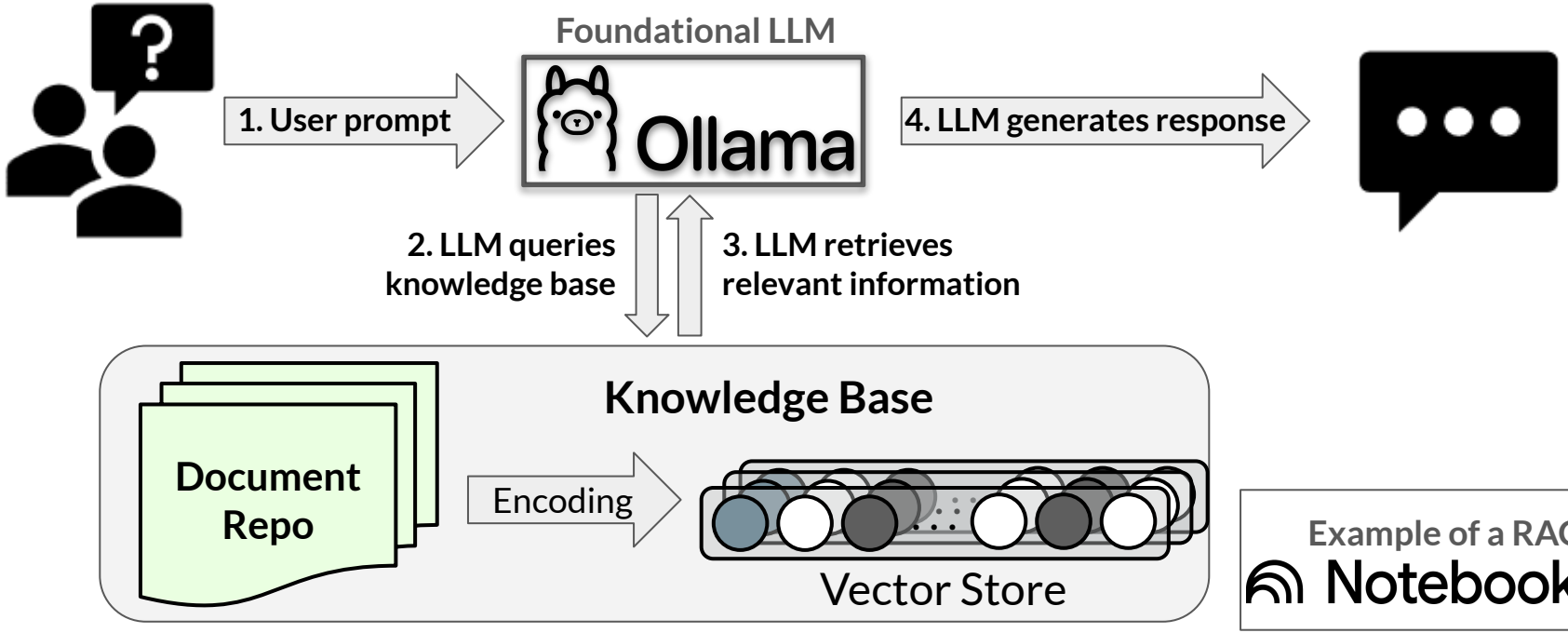
- Remove inappropriate uses
- Ensure safety

Alignment phases

Foundational Model

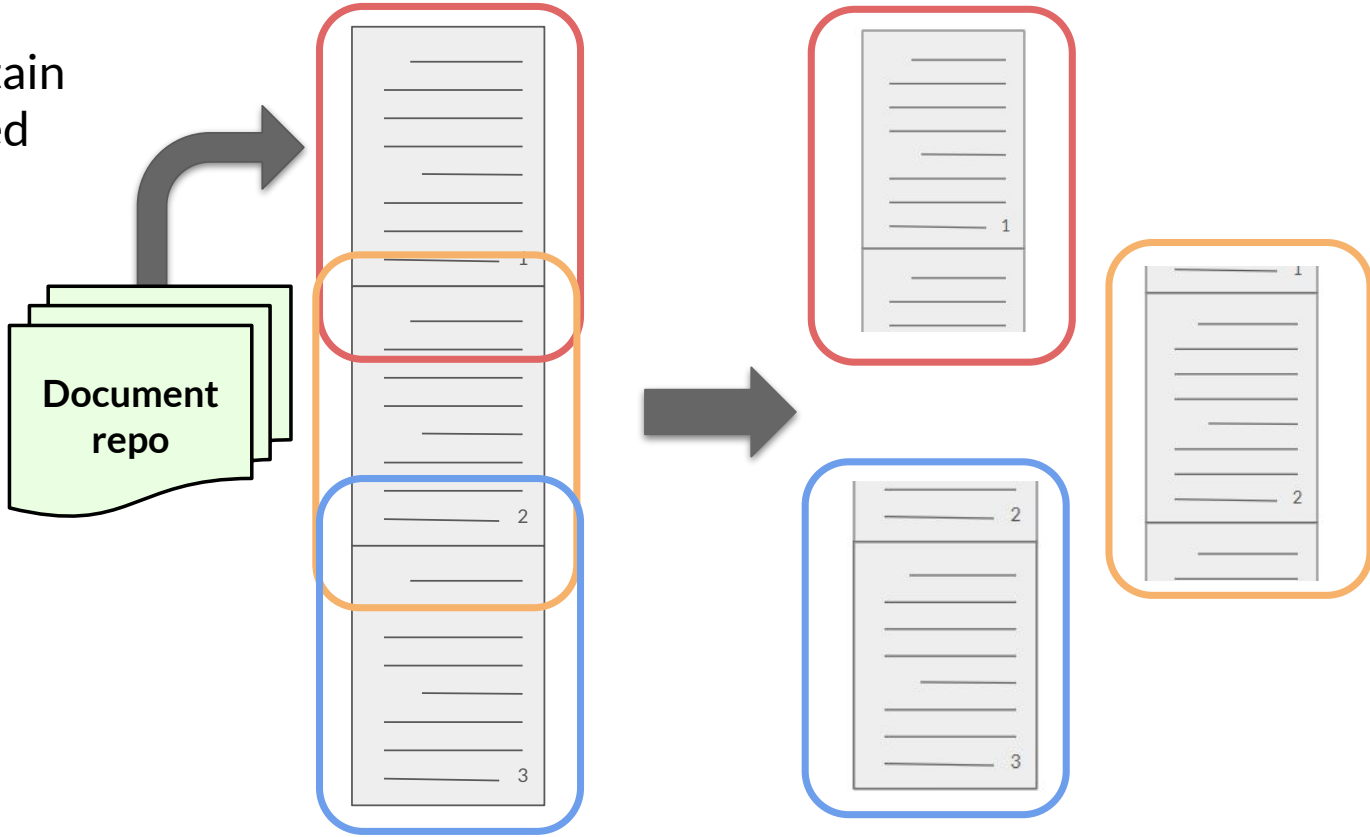
RAG / Knowledge Bases

- Retrieval augmented generation (RAG) models enable access to additional information to provide context and specificity in their responses.



Document Chunking

- Key step in generating knowledge bases.
- Overlap helps maintain context when divided between chunks



MSU AI Policy/Guidelines

GenAI Policy and Ethical Considerations for Researchers

References for Using Generative AI Technology And Services Responsibly in a Research Setting at Michigan State University

Patrick Bills, MSU ICER Research Software Engineer – March 2026

Introduction

What are we covering?

Using GenAI responsibly in an institutional research setting has several levels of consideration

- Institutional Policies and Compliance
- Personal Ethics
- Practical Considerations

MSU Researchers must balance these for fruitful use of GenAI in their work

My goal is to lightly cover these topics & point out highlights and references

I’m not a lawyer, policy-maker, or IT Services employee ... views are my own interpretation

Please use the contacts listed on the AI Guidance and policy websites listed in these slides for definitive information

MSU Institutional Policies

What is the context for Responsible GenAI on campus?

In order of specificity, these are MSU policies in which we work

- GenAI Ethical Standards for Research
- GenAI General Guidelines
- Institutional Data Policy
- MSU Acceptable Use Policy
- Intellectual Property Policies

Other research institutions have similar frameworks

MSU General Policies Governing GenAI Policy

Policies come from several sources on campus

Technology Policies: MSU IT Services

- https://tech.msu.edu/about/guidelines-policies/generative-ai/
- https://tech.msu.edu/technology/ai/tools/ contracts and agreements for vendors

Ethics Institute https://ethics.msu.edu/

- https://ethics.msu.edu/gen-ai

MSU Research

- https://research.msu.edu/compliance - see sections on research integrity and research security
- https://research.msu.edu/generative-ai

MSU Provost/President: https://ai.msu.edu

GenAI Policies: IT Services

"Interim Guidance on Data Uses and Risks of Generative AI"

- https://tech.msu.edu/about/guidelines-policies/generative-ai/
- March 2026:
  "*Currently, Microsoft Copilot with enterprise data protection (available for a cost to all MSU faculty, staff, and students when logged in with your MSU credentials) is the only MSU IT approved generative AI tool for use with the following data types:*"
  - Institutional data
  - Research data (with appropriate ownership or permissions)
  - FERPA data

GenAI Policies: MSU Research Data

MSU Current Policy:
Unpublished research data should not be used with GenAI tools
except for licensed enterprise version of co-pilot, and only with approval

- Entering confidential or sensitive information into third-party, non-MSU enterprise generative AI tools is strictly prohibited unless explicitly approved by MSU IT Information Security.
- Users are expected to become proficient in the use of digital tools and exercise caution when entering confidential or sensitive information into generative AI tools. They must review the MSU institutional data policy to understand the potential risks associated with generative AI tools and seek clarity to ensure that MSU-approved generative AI tools are authorized (or not) to handle such data (e.g., FERPA-protected, HIPAA-regulated, unpublished research).
- Where to go for assistance?
  - MSU Institutional Review Board
  - MSU IT Services Academic Technology

MSU Cloud Computing Policies Apply!

Policies governing cloud computing are relevant to AI services

https://tech.msu.edu/about/guidelines-policies/cloud-services-appropriate-use/

- When you sign up for a cloud-based service, you almost always agree to the terms of use.
- You are not legally able agree to terms when using MSU resources (systems, network, data).

“Only a small number of individuals at MSU (typically the highest-ranking executive officers) are authorized by the MSU Board of Trustees to enter into legal contracts on behalf of the university. When individual users without such signature authority wish to use a cloud service for their university work and accept a click-through agreement, they are individually responsible and personally liable for any legal liabilities resulting from the use of the cloud service.”

The link above has great considerations before signing up for any cloud service for personal use.

- Most GenAI services are cloud services, and many hide their terms.
  - For example, Perplexity AI terms are not on their front page: https://www.perplexity.ai/hub/legal/terms-of-service which sounds like they don’t store data, but the external APIs they use might do that. Don’t use MSU data with services like this.

GenAI Ethics

From the MSU AI Guidelines https://ai.msu.edu/guidelines

- **Collaboration** – We prioritize reciprocity and ethical partnerships, aiming to drive interdisciplinary innovation that benefits both our community and society at large.
- **Equity** – We are committed to maximizing accessibility and equity, actively dismantling biases and barriers related to socioeconomic status, disabilities and institutional boundaries to empower all members of our community.
- **Excellence** – We will ethically employ AI for institutional enhancement and societal betterment, supporting professional development and continuous policy refinement to elevate our standards in all facets of our work.
- **Integrity** – We commit to responsible AI use, adhering to all laws, acknowledging its limitations, ensuring privacy, security and unbiased data handling, while rigorously reviewing outputs to maintain our standard of honesty and trustworthiness.
- **Respect** – We pledge to utilize AI in ways that honor human dignity and foster a culture of safety and understanding, valuing human insight above all while acknowledging and mitigating AI biases to ensure a respectful, secure community.

Responsible GenAI: other ethical considerations

from Cloud Computing Training Materials – not specific to MSU

- **Fairness** helps ensure that AI systems do not discriminate against individuals or groups.
- **Safety** focuses on helping ensure that AI systems do not cause unintended harm to users or society.
- **Privacy** and security protect user data and prevent unauthorized access or misuse of AI systems.
- **Veracity** and robustness help ensure that AI systems produce reliable and consistent results, even in the face of unexpected inputs or changing environments.
- **Transparency** promotes openness about AI capabilities, limitations, and potential risks.
- **Explainability** helps humans understand how AI systems arrive at their decisions.
- **Controllability** allows for human oversight and intervention in AI systems when necessary.
- **Governance** provides a framework for implementing and enforcing responsible AI practices within an organization.

Responsible GenAI: mediating issues

Mediating Issues when using External LLMs

- Most applications of GenAI rely on externally created foundational large language models (LLMs)
- These are frequently proprietary, biased, and not transparent.
- It's your job to find systems that:
  - can identify where information came from (provide references)
  - control the source of information: knowledge base + prompt example in our first tutorial
  - are vetted and approved by your institution
- Privacy: use systems your institution has vetted to ensure privacy, review privacy policies. Many GenAI platforms (including Enterprise Azure and AWS) do not share or re-used data input
- Governance: follow institutional guidelines

Responsible GenAI: restrictions

Don’t put chatbots or other AI tools online

- You are not allowed to publish GenAI application that appear to be promoted by, or speak on behalf-of, your institution (MSU)
- Chatbots are frequently used in cloud computing tutorials but are especially problematic if they are
  - published with MSU logos or words
  - published as part of a laboratory's work
  - Don’t use public-domain data or use your own, private accounts or MSU Accounts.

If you want to experiment with building a chatbot, do not publish online, run from your laptop only.

Responsible GenAI: MSU Approved Tools

Details taken from an email sent to the MSU community by MSU IT Services
February 19, 2026

MSU-LICENSED AI TOOL	COST	ACCESS
Microsoft Copilot	No cost	copilot.microsoft.com
Microsoft 365 Copilot	Departmental purchase	techstore.msu.edu
OpenAI ChatGPT	Departmental purchase	techstore.msu.edu
Google Gemini	No cost	googleapps.msu.edu
Google NotebookLM	No cost	googleapps.msu.edu

Google Gemini and NotebookLM are now available at no cost to all students, faculty, and staff with an MSU Google account. These AI tools support tasks such as writing assistance, summarization, and organization. NotebookLM helps structure and explore notes using AI-powered insights. MSU IT is exploring the ability for users to purchase upgraded (Pro) versions of these tools. Access them with your MSU credentials at googleapps.msu.edu.

Microsoft Copilot Chat is available at no cost to students, faculty, and staff with an MSU Spartan 365 account. It supports tasks such as coding, writing, image generation, and general Q&A. Copilot uses a combination of AI models and ensures enterprise-level data protection. When accessed with MSU credentials, data entered remains secure and is not used to train the AI model. Access Copilot using your MSU credentials at copilot.microsoft.com.

Microsoft 365 Copilot integrates AI directly into Outlook email and core Microsoft apps including Word, Excel, Teams, PowerPoint, and content stored in SharePoint. It securely connects your information across Microsoft apps and connected systems to help you analyze, write, summarize, and present more efficiently.

M365 Copilot has recently dropped in price and is available only to faculty, staff, and students whose departments purchase licenses through techstore.msu.edu.

OpenAI ChatGPT is available as an enterprise application for faculty and staff. This version offers enhanced data protection and improved privacy terms. It requires departmental purchase and is currently classified as a Tier 3 application, which includes an annual IT Readiness form. Learn more about ChatGPT and purchase ChatGPT Edu at the MSU Tech Store.

Responsible GenAI: positive examples

*What **can** you do with GenAI?*

- create internal systems for use in your lab or research group
  - use MSU cloud accounts to create internal knowledge bases for searching
  - example: "*pathfinder - A Semantic Framework for Literature Review and Knowledge Discovery in Astronomy*" https://arxiv.org/abs/2408.01556
  - Many tutorials will show you how to do this
- Data pipelines
  - Can use MSU approved AI tools (co-pillot, OpenAI, or Gemini) with public input data
  - for use with unpublished research data, seek approval from MSU IT Services to leverage AWS data storage, analysis and GenAI tools
- Use Co-Pilot
- Use MSU ICER!
  - Open-source tools and LLMs (not proprietary)
  - data remains on campus and not restricted as it is for cloud service providers

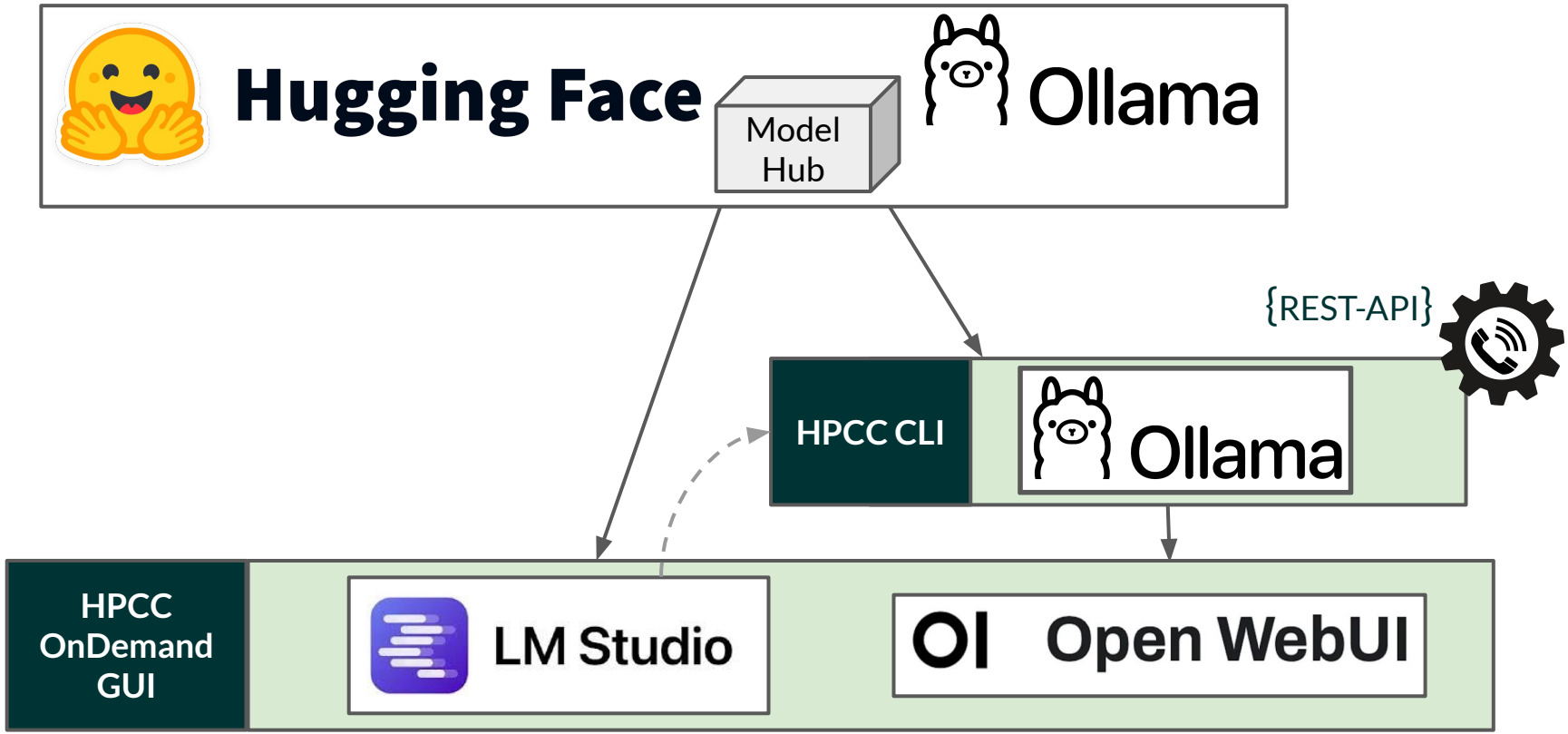
Summary

GenAI Policy and Guidlines for MSU, March 2026

- *We are users of MSU GenAI guidelines, not authors or policy makers*
- *Always check the websites listed in these slides for the latest guidance*
- *send questions About GenAI Policy and Guidelines for MSU to contacts for the provost, MSU Research and MSU IT*

AI Tools on the HPCC

AI Tools on the HPCC



AI Tools on the HPCC



Reference Data

Web Search

Knowledge Bases (RAG)

Chat

OpenWebUI

LMStudio

{REST-API}

Local Model Hub

Common

Personal /Group

Training Data

Fine Tuning

Ollama

```
>_
{"role": "system", "content": system_prompt}
{"role": "user", "content": example_input}
{"role": "assistant", "content": example_output}
{"role": "user", "content": user_prompt}
```

Tools

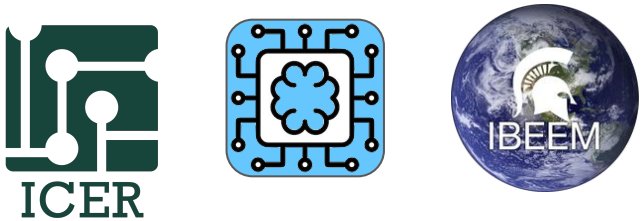
- Research Workflows
- Local LLM apps
- Agentic systems

- Curl
- Python (http/api libraries, langchain)

<HPCC OnDemand>

https://ondemand.hpcc.msu.edu

GenAI on the HPCC: Week 2



Julian Venegas
Institute for Cyber-Enabled Research

Kelly Kapsar
Institute for Biodiversity, Ecology, Evolution, & Macrosystems

Week 2 Outline

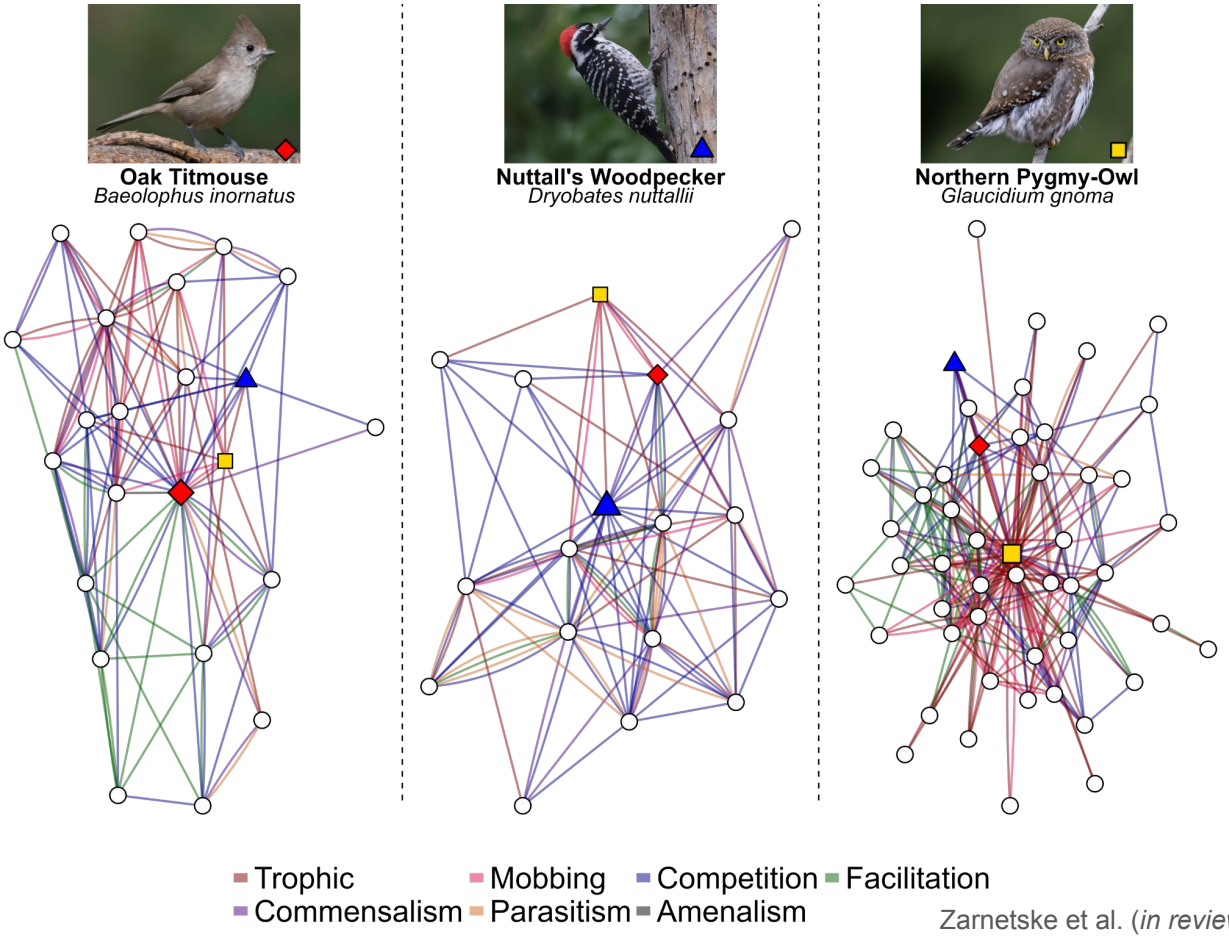
- Review and questions
  - Ollama demo
- Updates:
  - Open WebUI is (temporarily) down to fix the bug you all helped to identify next week!
  - LM Studio is functional
- Unstructured output example:
  - ICER docs bot (Jupyter notebook)
- Structured output example:
  - Intro to Avian MetaNetwork
  - Avian MetaNetwork (Jupyter notebook)

Use Cases

Use case: Avian Interaction Extraction

AvianMetaNet

- Understanding how networks of species interact with each other can help us to predict the effects of species loss on ecosystem functioning.



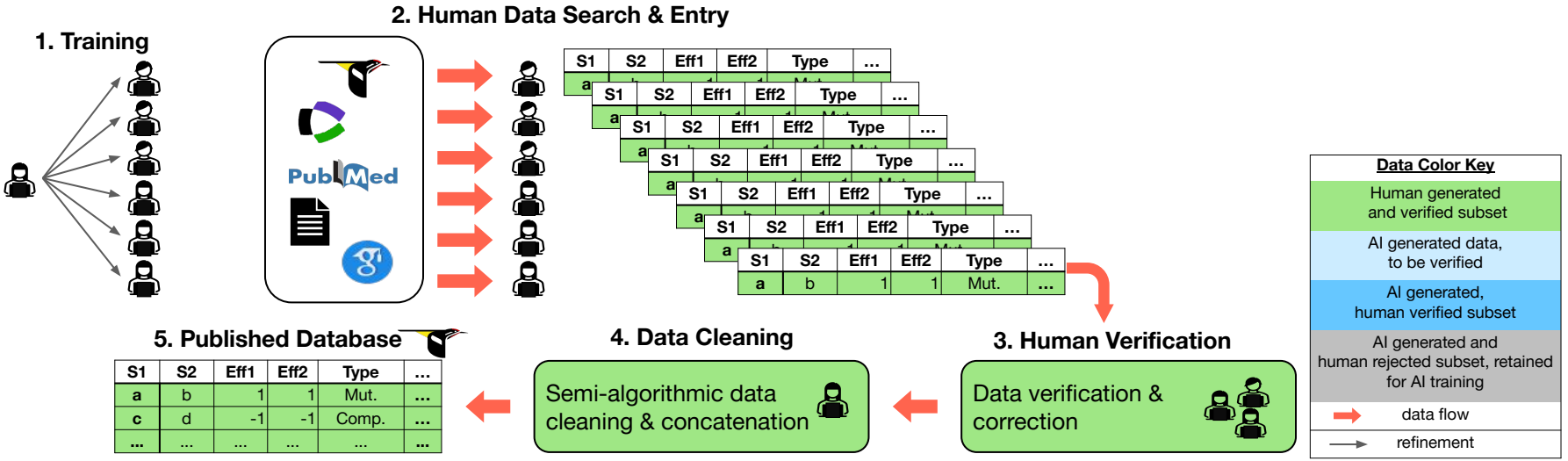
Zarnetske et al. (*in review*)

AvianMetaNet

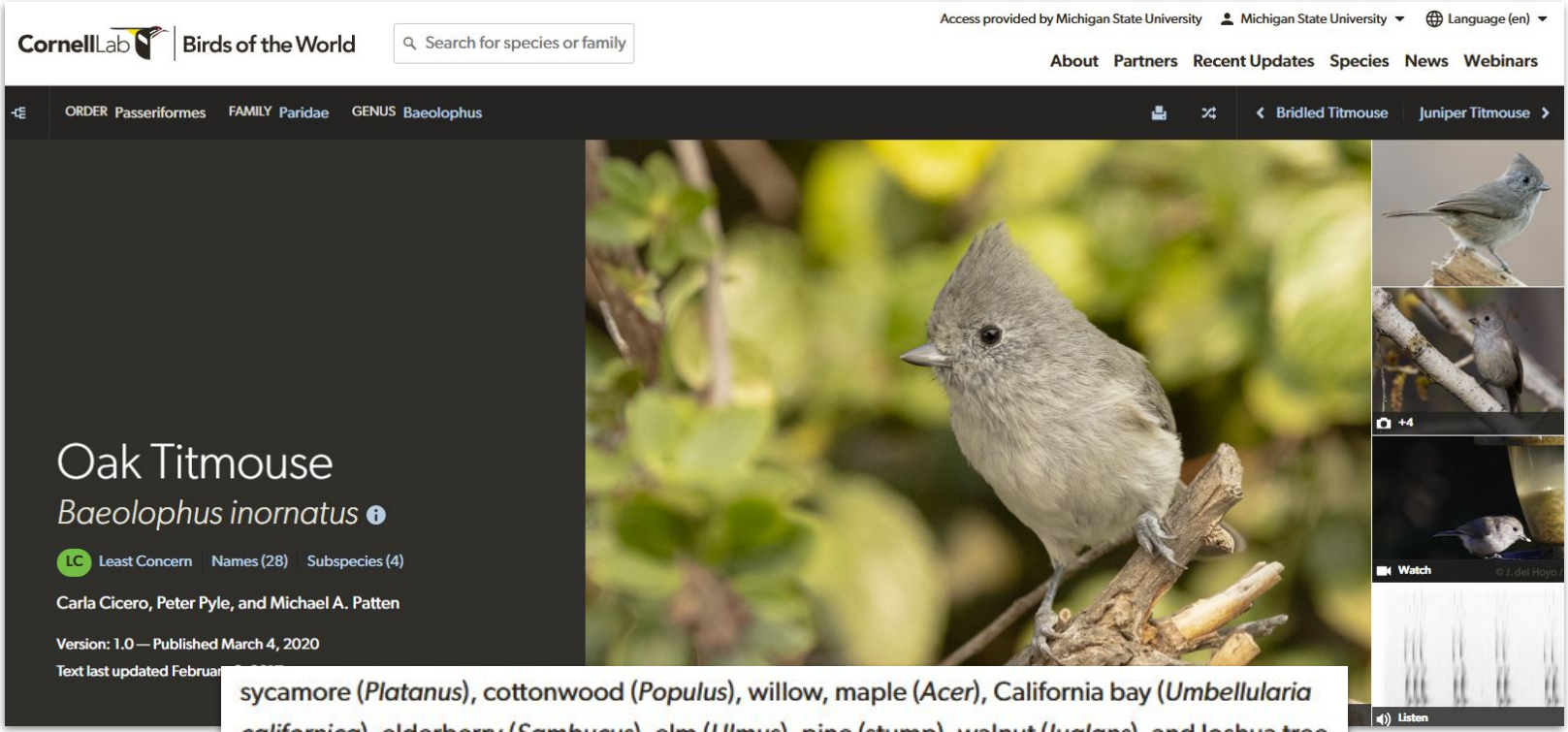
- **Knowledge gap:**
  - Lack of synthesized data sets on species interactions (e.g., predation, facilitation, commensalism)
- **Current approach:**
  - Systematic review of scientific literature and bird ecology databases
  - Manual coding of documented interactions by trained recorders
  - Semi-automated processing and integration of species records into

CornellLab | Birds of the World PubMed

AvianMetaNet: Manual workflow



Use case: Avian Interaction Extraction



sycamore (*Platanus*), cottonwood (*Populus*), willow, maple (*Acer*), California bay (*Umbellularia californica*), elderberry (*Sambucus*), elm (*Ulmus*), pine (stump), walnut (*Juglans*), and Joshua tree. Two nests recorded in scrub oak. Excavated holes used by Oak Titmouse include those dug by Nuttall's Woodpecker (*Picoides nuttallii*), Downy Woodpecker (*P. pubescens*), Ladder-backed Woodpecker (*P. scalaris*), Acorn Woodpecker (*Melanerpes formicivorus*), and Northern Flicker (*Colaptes auratus*).

Use case: Avian Interaction Extraction

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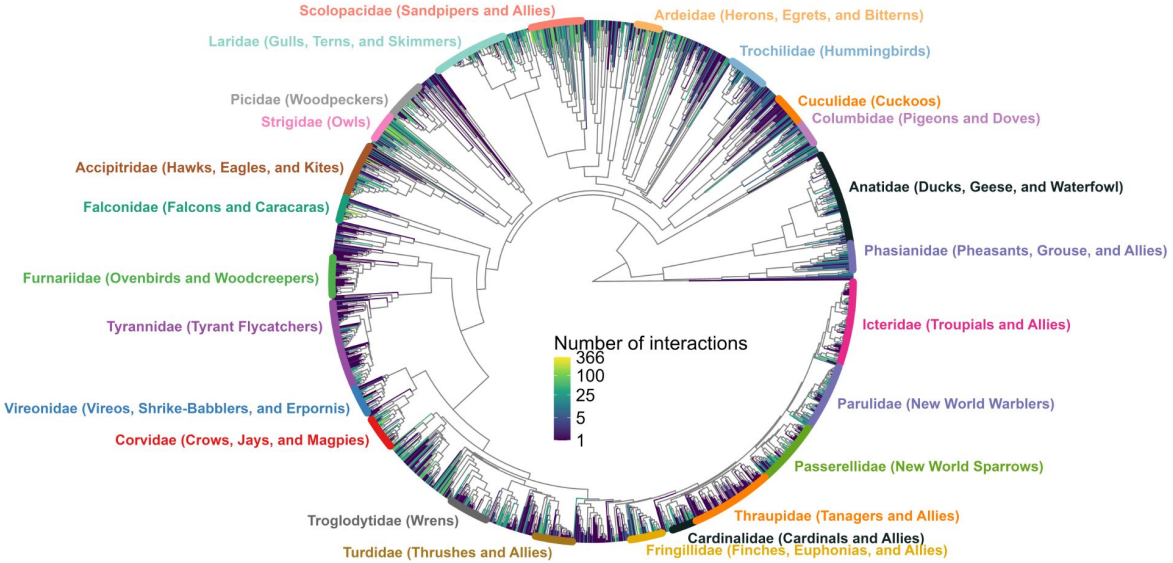
species1_common	species2_common	species1_scientific	species2_scientific	effect_sp1_on_sp2	effect_sp2_on_sp1	interaction
Oak Titmouse	Nuttall's Woodpecker	Baeolophus inornatus	Picoides nuttallii	0	1	commensalism

BOW_evidence	n_studies	notes	recorder	entry_date	encertain_interaction	entry_changes
strong	1	Nest Site: Microhabitat; Site Characteristics: Excavated holes used by Oak Titmouse include those dug by Nuttall's Woodpecker (Picoides nuttallii), Downy Woodpecker (P. pubescens), Ladder-backed Woodpecker (P. scalaris), Acorn Woodpecker (Melanerpes formicivorus), and Northern Flicker (Colaptes auratus).	PLZ	3/16/2022		

Use case: Avian Interaction Extraction

AvianMetaNet

- The Database:
  - 25,948 documented instances of interactions between 1,989 unique taxa representing ~14% of all bird species.



Zarnetske et al. (*in review*)

AvianMetaNet

- **Challenges:**
  - Time consuming, tedious, and expensive to manually collect and extract interaction data
  - Requires abstraction from written text (i.e., can't just Ctrl+F for keywords)
    - Changing species taxonomy
- **LLM-based approach:**
  - Use existing, manually extracted data as training/testing data for developing a retrieval augmented generation (RAG) model.

AvianMetaNet: AI-enhanced workflow

